

Name:		 MANIPAL UNIVERSITY JAIPUR	
Enrolment No:			
Summer/Makeup Examination, July- 2025 Faculty of Science, Technology and Architecture School of Computer Science and Engineering Department of Data Science B.Tech. Data Science			
Course Code: DS4145		Course: Social Network Analysis	
Time: 03 hrs.		Semester: VII	
		Max. Marks: 80	
Instructions: All questions are compulsory. Missing data, if any, may be assumed suitably. Calculator is allowed.			
SECTION A (Memory Based Questions)			
S.No.		Marks	CO
Q 1	In a social network, what do nodes and edges represent?	2	DS4145 .1
Q 2	In a _____ graph, edges have a direction, showing the flow from one node to another.	2	DS4145 .1
Q 3	The measure of how often a node appears on the shortest paths between other nodes is called _____ centrality.	2	DS4145 .2
Q 4	In a directed graph, edges: a) Have no orientation b) Point from one node to another in a specific direction c) Represent reciprocal relationships only d) Always form a cycle	2	DS4145 .2
Q 5	Which of the following is NOT a common network measure? a) Degree centrality b) Eigenvector centrality c) Page ranking d) Modularity	2	DS4145 .2
SECTION B (Concept Based Questions)			
Q 6	Consider a network of 3 nodes: A, B, and C, with the following edges: A→B; B→C; C→A Set the initial PageRank scores for each node to 1/3. Include teleportation with a damping factor of d=0.85 Compute the PageRank scores for one iteration and two iterations.	8	DS4145 .3
Q 7	A small, directed graph has 4 nodes: A, B, C, D with the following edges: A→B A→C ; B→C B→D; C→D Calculate the hub and authority scores for all nodes after two iterations. Assume initial scores are 1 for all nodes.	8	DS4145 .3
Q 8	A network consists of 5 nodes A,B,C,D,E and the following adjacency matrix: $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$	8	DS4145 .1

	Calculate the degree centrality, closeness centrality, and betweenness centrality of node C		
Q 9	Define influence in the context of social network analysis. How is influence propagated through a network, and what factors determine the strength and reach of influence? Provide examples of real-world scenarios where network influence plays a critical role	6	DS4145 .4
SECTION-C (Analytical Based Questions)			
Q 10	Consider a network with the following structure: Node A is connected to B by a directed edge and to C by an undirected edge. Node B is connected to D by an undirected edge. Node C is connected to E by a directed edge. Calculate the SimRank between nodes A and E, where directed and undirected links contribute differently to the similarity score. Assume a decay factor $C=0.8$ and that undirected edges contribute more to similarity.	10	DS4145 .3
Q 11	What are influence-related statistics in social networks, and how are they used to measure social influence within a network? Discuss some common metrics.	10	DS4145 .4
SECTION-D (Case Study/Application based Question)			
Q 12	a) What are some common techniques for anomaly detection in static graphs? b) Describe a use case where SimRank could be used to detect anomalous behavior in a social network. c) How would you apply PageRank to rank webpages in a search engine?	7+7+6	DS4145 .4